Ada (Barach) Lamba

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RESEARCH INTERESTS

AI Explainability, Speech and Sound Assessment, Human Behavioral Studies

EDUCATION

The Ohio State University Columbus, OH
PhD, Computer Science & Engineering Expected Graduation: May 2027
Advisor: Dr. Donald S. Williamson
Distinguished University Fellow

M.S., Computer Science & Engineering May 2026
Advisor: Dr. Donald S. Williamson
Quinlan Teaching Award 2021

B.S., Computer Science & Engineering May 2020
Magna Cum Laude

PUBLICATIONS

- Kibria, I., **Lamba, A.**, Williamson, D., *A Generalization Strategy for Speech Quality Prediction: From Domain-Specific to Unified Datasets*. Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2026.
- Beardsley, V., Xiong, C., **Lamba, A.**, Bond, M. D., *Carapace: Static-Dynamic Information Flow Control in Rust.* Proc. ACM Prog. Lang. 9 (OOPSLA), 2025.
- **Lamba, A.**, Taylor, M., Beardsley, V., Bambeck, J., Bond, M.D., Lin, Z., *Cocoon: Static Information Flow Control in Rust.* Proc. ACM Prog. Lang. 8 (OOPSLA), 2024.
- Kecskemety, K., **Barach, A.**, Jenkins, C., Gunawardena, S., *Using Programming Concept Inventory Assessments: Findings in a First-year Engineering Course*. ASEE Annual Conference, 2021.
- **Barach, A.**, Jenkins, C., Gunawardena, S., Kecskemety, K., *MCS1: A MATLAB Programming Concept Inventory for Assessing First-year Engineering Courses.* ASEE Annual Conference, 2020.

(Under review)
- **Lamba, A.**, Williamson, D. S., *How Frequency Band Importance Affects Neural Network Predictions and Human Perception for Speech Quality Assessment.* Proc. Interspeech.

INDUSTRY EXPERIENCE

Handshake AI
MOVE Fellow, STEM Specialist Apr 2025 – Present
- Leveraging domain-specific knowledge to assist clients in training large language models

Battelle Memorial Institute Columbus, OH
Graduate Data Science Intern May 2020 – Jul 2020
- Developed visualization software for programmable FPGAs for use in side-channel attack analysis
- Refined machine learning algorithms for solving the Kaggle Acquired Valued Shoppers Challenge

MIT Lincoln Laboratory Lexington, MA
Summer Research Program Intern May 2019 – Jul 2019
- Developed auto-generation of data visualization for data analysis suite
- Obtained Interim Secret Clearance

TeamDynamix Columbus, OH
Engineering Intern May 2018 – Jul 2018

- Maintained and improved work management software as a full-time, full-stack software engineer, focusing on SQL and database queries, C# console applications, and web development

RESEARCH EXPERIENCE

The Ohio State University, ASPIRE Group, Department of Computer Science & Engineering Columbus, OH
Graduate Researcher, advised by Dr. Donald S. Williamson Jan 2023 – Present

- Investigating which features humans and neural networks focus on during prediction to examine possible causes for lack of correlation between human and network predictions.
- Using explainability methods to determine which features of an audio signal neural networks are focusing on to predict speech quality.
- Conducting a large scale human subject study to produce one of the first large data sets for speech intelligibility on which we train a neural network to predict intelligibility.

The Ohio State University, Department of Computer Science & Engineering Columbus, OH
Graduate Researcher, advised by Dr. Michael D. Bond, Dr. Zhiqiang Lin May 2021 – Apr 2025

- Developed a library for static information flow analysis in Rust programs.

The Ohio State University, FYRE, Engineering Education Department Columbus, OH
Undergraduate Researcher, advised by Dr. Krista Kecskemety Feb 2019 – May 2020

- Replicated, proctored, and validated a concept inventory for foundational computer science topics in MATLAB

The Ohio State University, Driving Simulation Laboratory Columbus, OH
Undergraduate Researcher, advised by Dr. Thomas Kerwin Nov 2018 – Feb 2019

- Conducted human subject research measuring driver behavior
- Developed and refined data analysis software

TEACHING EXPERIENCE

The Ohio State University, Department of Computer Science & Engineering Columbus, OH
Graduate Teaching Associate 2021 – 2026

- Taught course laboratory sections for Programming in Java and Data Structures in Java course
- Managed a team of undergraduate graders

LEADERSHIP AND PROFESSIONAL INVOLVEMENT

Member, IEEE's Signal Processing Society Feb 2026 – Present
Mental Wellness Committee, Department of Computer Science & Engineering (OSU) Aug 2021 – May 2025
Co-leader and Treasurer, Women in Cybersecurity Club (OSU) Jul 2021 – May 2022
Elected Delegate, Council of Graduate Students (OSU) Oct 2020 – Jul 2021
Member, Society of Women Engineers Aug 2019 – May 2022

SKILLS

Programming Languages: Python, C/C++/C#, Java, SQL, MATLAB, Rust
Machine Learning & Data Science: NumPy, Pandas, PyTorch, scikit-learn, TensorFlow
Parallel Computing: CUDA, SLURM